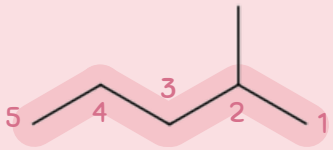
# PART A · STRUCTURE → NAME
## AO1 · Foundation · $C_6H_{14}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The longest chain is 5 C, so the parent is pentane. Two 5-C paths exist (the one through the left end of the zigzag, and the one that runs through the upward methyl). Each leaves exactly one methyl as a branch, so they lead to the same name.

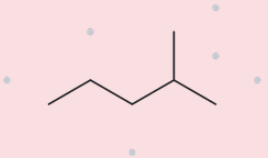
**Numbering:** Numbering left to right puts the methyl on C4. Numbering right to left puts it on C2. The first point of difference is 2 < 4, so number from the right.

**Substituents:** One methyl on C2, so the name is 2-methylpentane.

### 3 · MY NAME

2-methylpentane

### 4 · THE ALKANE CHECK



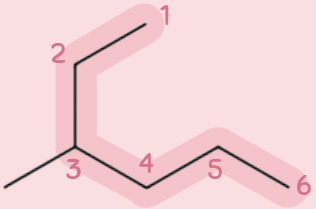
Returned Name: 2-methylpentane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO2 · Foundation · $C_7H_{16}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The longest chain is 6 C, so the parent is hexane. Going through the methyl gives only 5 C (methyl, C3, C4, C5, C6) or 4 C (methyl, C3, C2, C1), so the 6-C chain is the only longest one.

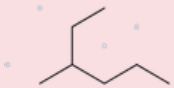
**Numbering:** Numbering from the upper-right end puts the methyl on C3. Numbering from the lower-right end puts it on C4. The first point of difference is 3 < 4, so number from the upper-right end.

**Substituents:** One methyl on C3, so the name is 3-methylhexane.

### 3 · MY NAME

3-methylhexane

### 4 · THE ALKANE CHECK



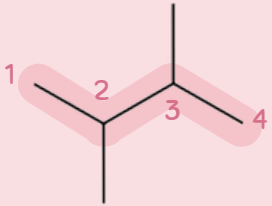
Returned Name: 3-methylhexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO3 · Foundation · $C_6H_{14}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The longest chain is 4 C, so the parent is butane. Several 4-C paths exist, since you can run through either the upward or the downward methyl instead of the end carbons. Each one leaves two methyl groups as branches, so they all lead to the same name.

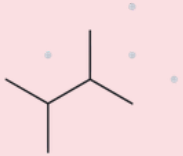
**Numbering:** Numbering left to right gives methyls on C2 and C3. Numbering right to left also gives C2 and C3. There is no point of difference, so either direction works.

**Substituents:** methyl on C2 and methyl on C3, so the name is 2,3-dimethylbutane.

### 3 · MY NAME

2,3-dimethylbutane

### 4 · THE ALKANE CHECK



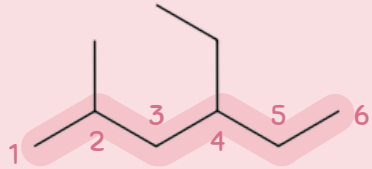
Returned Name: 2,3-dimethylbutane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO4 · Developing · $C_9H_{20}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The longest chain is 6 C, so the parent is hexane. Several 6-C paths exist, since you can run through the ethyl branch or the methyl branch instead of the zigzag ends. Each one leaves one methyl and one ethyl as branches, so they all lead to the same name. The path through the ethyl and the zigzag's right end is only 5 C, so it is not a candidate.

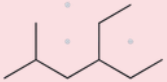
**Numbering:** Left to right gives substituents at 2,4. Right to left gives 3,5. The first point of difference is 2 < 3, so number from the left.

**Substituents:** Methyl on C2 and ethyl on C4. List them alphabetically (ethyl before methyl): 4-ethyl-2-methylhexane.

### 3 · MY NAME

4-ethyl-2-methylhexane

### 4 · THE ALKANE CHECK



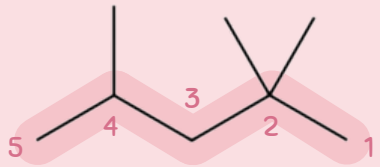
Returned Name: 4-ethyl-2-methylhexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
# AO5 · Developing · $C_8H_{18}$

## 1 · NUMBERED PARENT




## 2 · NAMING TRAIL

**Parent:** The longest chain is 5 C, so the parent is pentane. Several 5-C paths exist, since you can run through either upward methyl on the quaternary carbon instead of the right-hand end. Each one leaves three methyls as branches, so they all lead to the same name.

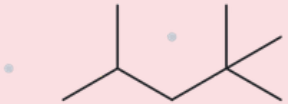
**Numbering:** Left to right gives methyls at 2,4,4. Right to left gives 2,2,4. The first point of difference is 2 < 4 (second locant), so number from the right.

**Substituents:** Two methyls on C2 and one on C4. Use "tri" and repeat the locant for the two methyls on the same carbon: 2,2,4-trimethylpentane.

## 3 · MY NAME

2,2,4-trimethylpentane

## 4 · THE ALKANE CHECK



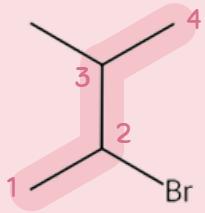
Returned Name: 2,2,4-trimethylpentane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO6 · Developing · $C_5H_{11}Br$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The longest chain is 4 C, so the parent is butane. Two 4-C paths exist, and each leaves one methyl as a branch, so they lead to the same name. Br is a substituent, not part of the chain.

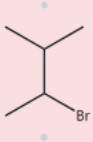
**Numbering:** Left to right gives bromo on C2 and methyl on C3. Right to left gives methyl on C2 and bromo on C3. Both sets are 2,3, so the tie goes to alphabetical order: bromo comes first, so it gets C2. Number from the lower-left end.

**Substituents:** Bromo on C2 and methyl on C3, so the name is 2-bromo-3-methylbutane.

### 3 · MY NAME

2-bromo-3-methylbutane

### 4 · THE ALKANE CHECK



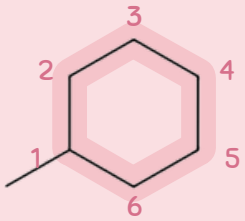

Returned Name: 2-bromo-3-methylbutane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO7 · Developing · $C_7H_{14}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The ring has 6 C and the only chain outside it is a 1-C methyl, so the ring is the parent: cyclohexane.

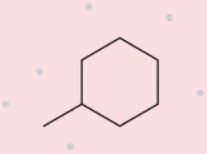
**Numbering:** With one substituent, it takes C1 by default, so no locant is written.

**Substituents:** One methyl, so the name is methylcyclohexane (not 1-methylcyclohexane).

### 3 · MY NAME

methylcyclohexane

### 4 · THE ALKANE CHECK



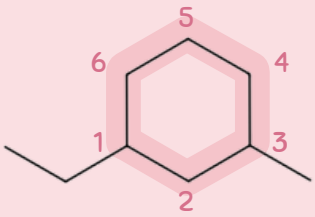
Returned Name : methylcyclohexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO8 · Challenge · $C_9H_{18}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The ring has 6 C and the largest substituent is only 2 C, so the ring is the parent: cyclohexane.

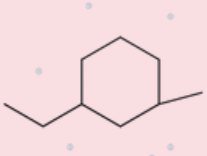
**Numbering:** The two substituents are 1,3 to each other whichever way you go, so the locants tie at 1,3. The tie goes to alphabetical order: ethyl comes before methyl, so ethyl gets C1 and methyl gets C3.

**Substituents:** Ethyl on C1 and methyl on C3, listed alphabetically: 1-ethyl-3-methylcyclohexane.

### 3 · MY NAME

1-ethyl-3-methylcyclohexane

### 4 · THE ALKANE CHECK



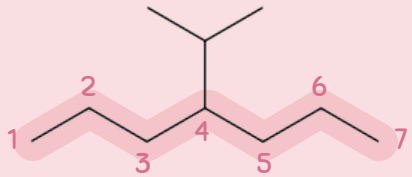
Returned Name: 1-ethyl-3-methylcyclohexane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
## AO9 · Challenge · $C_{10}H_{22}$

### 1 · NUMBERED PARENT




### 2 · NAMING TRAIL

**Parent:** The longest chain is 7 C, so the parent is heptane. Going up through the isopropyl gives at most 6 C ($CH_3$, CH, C4, C5, C6, C7), so the 7-C chain is the longest.

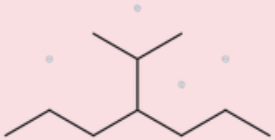
**Numbering:** The branch is at C4 from either end, so there is no point of difference and either direction works.

**Substituents:** The branch is a 3-carbon group attached through its middle carbon. Longest chain in the branch starting at the attachment carbon: 2 C, so the base is ethyl. The attachment carbon is C1 of that ethyl, and it carries one methyl, so the group is 1-methylethyl. Because the substituent name is complex, it goes in parentheses with the locant for the main chain: 4-(1-methylethyl)heptane.

### 3 · MY NAME

4-(1-methylethyl) heptane

### 4 · THE ALKANE CHECK



Returned Name: 2-methyl-3-propylhexane

MIS-MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART A · STRUCTURE → NAME
# A10 · Challenge · $C_7H_{14}$

## 1 · NUMBERED PARENT




## 2 · NAMING TRAIL

**Parent:** The ring has 3 C and the chain has 4 C. Current IUPAC recommendations prefer the ring as the parent, so the parent is cyclopropane.

**Numbering:** With one substituent, it takes C1, so no locant is written.

**Substituents:** One unbranched 4-carbon group, butyl, so the name is butylcyclopropane.

## 3 · MY NAME

butylcyclopropane

## 4 · THE ALKANE CHECK



Returned Name: butylcyclopropane

MATCH

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

# PART B · NAME → STRUCTURE
# BO1 · Foundation · $C_6H_{14}$

## 1 · NAME DECODED

**Parent:** pentane, a 5-carbon chain.

**Substituents:** methyl on C2.

**Expected formula:** 5+1 = 6 C. For an open-chain alkane, $C_nH_{2n+2}$ gives 2(6) + 2 = 14 H, so $C_6H_{14}$.

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_6H_{14}$

- 3 $CH_3$ (C1, C5 and the methyl branch) = 9 H
- 2 $CH_2$ (C3, C4) = 4 H
- 1 CH (C2) = 1 H
- Total = 14 H

**Open chain:** $C_nH_{2n+2} = C_6H_{14}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

2-methylpentane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO2 · Foundation · $C_7H_{16}$

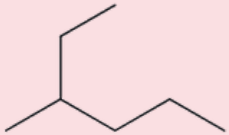
## 1 · NAME DECODED

**Parent:** hexane, a 6-carbon chain.

**Substituents:** methyl on C3.

**Expected formula:** 6 + 1 = 7 C. For an open-chain alkane, $C_nH_{2n+2}$ gives 2(7) + 2 = 16 H, so $C_7H_{16}$.

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_7H_{16}$

- 3 $CH_3$ (C1, C6 and the methyl branch) = 9 H
- 3 $CH_2$ (C2, C4, C5) = 6 H
- 1 CH (C3) = 1 H
- Total = 16 H

**Open chain:** $C_nH_{2n+2} = C_7H_{16}$

## 4 · BACK CHECK

Returned by The Alkane:

3-methylhexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO3 · Foundation · $C_6H_{14}$

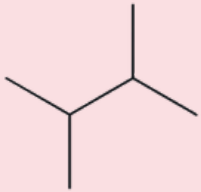
## 1 · NAME DECODED

**Parent:** butane, a 4-carbon chain.

**Substituents:** methyl on C2 and methyl on C3 (two methyls, so "di").

**Expected formula:** 4 + 2 = 6 C. For an open-chain alkane, $C_nH_{2n+2}$ gives 2(6) + 2 = 14 H, so $C_6H_{14}$.

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_6H_{14}$

- 4 $CH_3$ (C1, C4 and the two methyl branches) = 12 H
- 2 CH (C2, C3) = 2 H
- Total = 14 H

**Open chain:** $C_nH_{2n+2} = C_6H_{14}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

2,3-dimethylbutane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO4 · Developing · $C_9H_{20}$

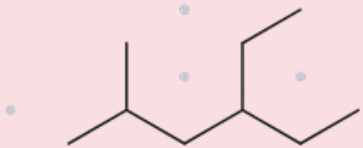
## 1 · NAME DECODED

**Parent:** hexane, a 6-carbon chain.

**Substituents:** methyl on C2; ethyl on C4 (listed alphabetically, ethyl before methyl).

**Expected formula:** 6 + 1 + 2 = 9 C. For an open-chain alkane, $C_nH_{2n+2}$ gives 2(9) + 2 = 20 H, so $C_9H_{20}$.

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_9H_{20}$

- 4 $CH_3$ (C1, C6, the methyl branch, the ethyl's end carbon) = 12 H
- 3 $CH_2$ (C3, C5, the ethyl's first carbon) = 6 H
- 2 CH (C2, C4) = 2 H
- Total = 20 H

**Open chain:** $C_nH_{2n+2} = C_9H_{20}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

4-ethyl-2-methylhexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BOS · Developing · $C_8H_{18}$

## 1 · NAME DECODED

**Parent:** pentane, a 5-carbon chain.

**Substituents:** two methyls on C2 and one methyl on C4 (three methyls, so "tri").

**Expected formula:** 5 + 3 = 8 C. For an open-chain alkane, $C_nH_{2n+2}$ gives 2(8) + 2 = 18 H, so $C_8H_{18}$.

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_8H_{18}$

- 5 $CH_3$ (C1, C5 and the three methyl branches) = 15 H
- 1 $CH_2$ (C3) = 2 H
- 1 CH (C4) = 1 H
- 1 C with no H (C2, quaternary) = 0 H
- Total = 18 H

**Open chain:** $C_nH_{2n+2} = C_8H_{18}$ ✓
**Carbon count:** 5 + 1 + 1 + 1 = 8 C ✓

## 4 · BACK CHECK

Returned by The Alkane:

2,2,4-trimethylpentane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO6 · Developing · $C_5H_{11}Br$

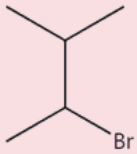
## 1 · NAME DECODED

**Parent:** butane → a 4-carbon chain.

**Substituents:** bromo on C2; methyl on C3.

**Expected formula:** 4 + 1 = 5 C; one chain → $C_5H_{12}$; Br replaces one H → $C_5H_{11}Br$

## 2 · BOND-LINE STRUCTURE (THE ALKANE)




## 3 · FORMULA CHECK

**Counted from the drawing:** $C_5H_{11}Br$

- 3 $CH_3$ (C1, C4, the methyl branch) = 9 H
- 2 CH (C2 and C3) = 2 H
- Total = 11 H, plus 1 Br

**Open chain:** $C_nH_{2n+2}$ = $C_5H_{12}$; minus one H for Br = $C_5H_{11}Br$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

2-bromo-3-methylbutane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO7 · Developing · $C_7H_{14}$

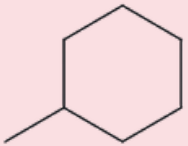
## 1 · NAME DECODED

**Parent:** cyclohexane → a 6-carbon ring.

**Substituents:** methyl on C1 (the locant is not written, since there is only one substituent).

**Expected formula:** 6 + 1 = 7 C; one ring → $C_6H_{12}$; the methyl replaces one H → $C_7H_{14}$

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_7H_{14}$

- 1 CH (C1) = 1 H
- 5 $CH_2$ (C2 to C6) = 10 H
- 1 $CH_3$ (methyl) = 3 H
- Total = 14 H

**One ring:** $C_nH_{2n}$ = $C_6H_{12}$; the methyl replaces one H and adds $CH_3$, so $C_6H_{11}$ + $CH_3$ = $C_7H_{14}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

methylcyclohexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO8 · Challenge · $C_9H_{18}$

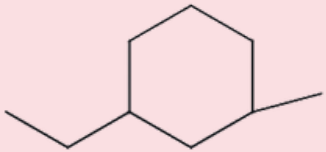
## 1 · NAME DECODED

**Parent:** cyclohexane → a 6-carbon ring.

**Substituents:** ethyl on C1; methyl on C3 (listed alphabetically).

**Expected formula:** 6 + 2 + 1 = 9 C; one ring → $C_6H_{12}$; ethyl and methyl each replace one H → $C_9H_{18}$

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_9H_{18}$

- 2 ring CH (C1, C3) = 2 H
- 4 ring $CH_2$ (C2, C4, C5, C6) = 8 H
- 1 ethyl $CH_2$ = 2 H
- 2 $CH_3$ (ethyl end, methyl) = 6 H
- Total = 18 H

**One ring:** $C_nH_{2n} = C_6H_{12}$; minus two H for the substituents = $C_6H_{10}$, plus $C_2H_5$ and $CH_3$ = $C_9H_{18}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

1-ethyl-3-methylcyclohexane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# BO9 · Challenge · $C_{10}H_{22}$

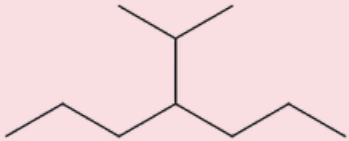
## 1 · NAME DECODED

**Parent:** heptane → a 7-carbon chain.

**Substituents:** 1-methylethyl (isopropyl) on C4.

**Expected formula:** 7 + 3 = 10 C; open chain → $C_7H_{16}$; the branch replaces one H → $C_{10}H_{22}$

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_{10}H_{22}$

- 4 $CH_3$ (C1, C7, two isopropyl methyls) = 12 H
- 4 $CH_2$ (C2, C3, C5, C6) = 8 H
- 2 CH (C4, isopropyl CH) = 2 H
- Total = 22 H

**Open chain:** $C_nH_{2n+2} = C_{10}H_{22}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

2-methyl-3-propylhexane

NOT same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded

# PART B · NAME → STRUCTURE
# B10 · Challenge · $C_7H_{14}$

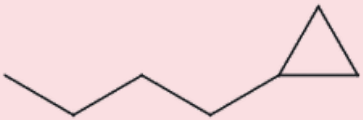
## 1 · NAME DECODED

**Parent:** cyclopropane → a 3-carbon ring.

**Substituents:** butyl on C1 (the locant is not written, since there is only one substituent).

**Expected formula:** 3 + 4 = 7 C; one ring → $C_3H_6$; butyl replaces one H → $C_7H_{14}$

## 2 · BOND-LINE STRUCTURE (THE ALKANE)



## 3 · FORMULA CHECK

**Counted from the drawing:** $C_7H_{14}$

- 1 ring CH (C1) = 1 H
- 2 ring $CH_2$ (C2, C3) = 4 H
- 3 chain $CH_2$ = 6 H
- 1 $CH_3$ = 3 H
- Total = 14 H

**One ring:** $C_nH_{2n} = C_3H_6$; butyl replaces one H, so $C_3H_5$ + $C_4H_9$ = $C_7H_{14}$ ✓

## 4 · BACK CHECK

Returned by The Alkane:

butylcyclopropane

Same as the given name

☑ parent length matches ☑ substituent positions match ☑ formula agrees ☑ back-check recorded